

202

Donnerstag 10. April
1912

Days

$$V = 40\text{ m}^3 = 0,4\text{ m}^3$$

$$R = 0,1 \cdot 10^6 \Omega$$

$$P \approx 0.5 \cdot 10^6 \text{ Pa}$$

Q-2

~10.2

Heinrich

$$P_0 V = n R T_0, \quad T_0 = \frac{P_0 V}{n R} =$$

$$\begin{array}{r} 6000 \\ 2714 \end{array}$$

$$PV = nRT; T = \frac{PV}{nR}$$

2000
2,1940

$$\Delta U = nC_v \Delta T = \frac{5}{2} n R \Delta T = \frac{5}{2} \cdot 1 \text{ mol} \cdot \frac{8.314 \text{ J}}{\text{mol} \cdot \text{K}} \cdot 10 \text{ K} = 207.85 \text{ J}$$

$$Q = \Delta U + W$$

Both on V-coast, re $W=0$, $\frac{1}{2} \frac{dW}{dt}$

$$Q = 1250 \text{ Nm} = 12.5 \text{ kNm}$$

6.6. 90.2.21

24

$$m = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

AT 242

$P^2 = 19.56$

$$= \log \left(\frac{1}{2} \right) = -1$$

$$\frac{21.03 \frac{\text{m}}{\text{h}}}{11.15} = 1.886 \text{ m/h}$$

$$V_1 = \frac{K_1}{r} \quad V_2 = \frac{K_2}{r}$$

$$\Delta V = \frac{nRT_1 + nRT_2 - nRT_1}{P} = \frac{nR\Delta T}{P}$$

Fig. 2. $W = 1$.

W. E. B. D.

مقامہ

2000

Q. 10

54-0

Allen

$$V \in C^{1,1}$$

1972-1973

$$\frac{104}{100} = 1.04$$

0	0
---	---

1. 2

$$\frac{1}{2} R =$$

100

W. Allen

1) $W = P \Delta V = \frac{P}{\gamma} \left(\frac{\gamma}{\gamma-1} \right) \Delta T$
 $W = 6,3125 \cdot 10^5 \cdot 2,5 \cdot 10^{-3} \approx 1,578 \text{ J}$
 Answer: $\approx 1,58 \text{ J}$

2) $Q = n C_V \Delta T$
 $Q = 1,578 \text{ J}$
 $\Delta T = ?$
 Answer: $\Delta T = 10,14 \text{ K}$
 Answer: $\Delta T = 10,14 \text{ K}$
 Answer: $\Delta T = 10,14 \text{ K}$

3) $Q = n C_V \Delta T = \frac{5}{2} R n \Delta T = \frac{5}{2} \cdot 8,314 \cdot 2 \cdot 10,14 \approx 850,9 \text{ J}$
 $\approx 850,9 \text{ J} \approx 1 \text{ kJ}$
 Answer: 1 kJ

4) $Q = n C_V \Delta T = \frac{5}{2} R n \Delta T = \frac{5}{2} \cdot 8,314 \cdot 2 \cdot 10,14 \approx 850,9 \text{ J}$
 $\approx 850,9 \text{ J} \approx 1 \text{ kJ}$
 Answer: 1 kJ

1) $V_0 = 0,001 \text{ m}^3$
 $P = 10^6 \text{ Pa} = \text{const}$
 $V_1 = 2V_0 = 0,002 \text{ m}^3$
 $Q = ?$
 Answer: $\Delta T = 10,15 \text{ K}$
 Answer: $\Delta T = 10,15 \text{ K}$
 Answer: $\Delta T = 10,15 \text{ K}$

2) $P V_0 = n R T_0$
 $T_0 = \frac{P V_0}{n R}$
 $3) Q = n C_V \Delta T = \frac{5}{2} R n \Delta T = \frac{5}{2} P \Delta V = \frac{5}{2} P (V_1 - V_0) = \frac{5}{2} \cdot 10^6 \cdot 0,001 = 2,5 \cdot 10^5 \text{ J} = 250 \text{ kJ}$
 Answer: 250 kJ

Answer: 250 kJ
 Answer: 250 kJ
 Answer: 250 kJ

1) $P = 10^6 \text{ Pa}$
 $V = 0,001 \text{ m}^3$
 $n = 1,1 \text{ mol}$
 $Q = 33,1 \text{ J}$
 $T_1 = 4,2 \text{ K}$
 $T_2 = 33,1 \text{ K}$
 $n, n' = ?$
 $n, n' = ?$
 Answer: $n, n' = ?$

Answer: $n, n' = ?$